The Influence of Oleic Acid Concentration on the Growth Kinetics and Size Distribution of CdSe Quantum Dots

Krishna Madhur Akella[1] and Dhruvi Rathore[1]

[1]Department of Electrical and Systems Engineering, University of Pennsylvania

Abstract

Colloidal semiconductor nanocrystals, specifically cadmium selenide (CdSe) quantum dots (QDs), exhibit size-tunable optoelectronic properties dictated by quantum confinement. The kinetic growth and thermodynamic stability of these nanocrystals are heavily governed by capping ligands during synthesis. In this study, we investigate the influence of oleic acid concentration on the growth kinetics and size distribution of CdSe QDs. By reducing the standard oleic acid volume in the cadmium precursor from 0.6 mL to 0.3 mL, we hypothesize a lowered steric energy barrier for monomer attachment. UV-Vis absorption spectra revealed an accelerated red-shift of the first exciton peak from 424.30 nm to 518.95 nm over 9.87 minutes (592.24 s), corresponding to a rapid physical diameter increase from 1.76 nm to 2.55 nm. However, fluorescence spectroscopy and Scanning Transmission Electron Microscopy (STEM) demonstrate a thermodynamic trade-off: the ligand deficiency induces Ostwald ripening and compromises colloidal stability, yielding a broader emission full-width at half-maximum (FWHM) that increased to 51.68 nm in the later stages of growth, alongside increased nanocrystal aggregation. This report quantifies the critical balance between reaction kinetics and thermodynamic control in nanomaterial synthesis.

1 Introduction

In this quantum confinement regime, the continuous energy bands of the bulk material transition into discrete atomic-like energy levels, allowing for precise tuning of the optical bandgap through structural sizing [1,2]. Cadmium selenide (CdSe) quantum dots (QDs) serve as an archetypal system for investigating these phenomena, offering highly tunable emission across the visible spectrum that is essential for next-generation optoelectronic devices [3].

The synthesis of highly monodisperse colloidal QDs relies on a delicate balance between nucleation and growth phases, which are fundamentally controlled by the thermodynamic and kinetic environment of the reaction [4]. Capping ligands, acting as surfactants, play a dual role in this process: they provide colloidal stability to prevent uncontrolled aggregation and act as a steric barrier regulating the diffusion and attachment of unreacted monomers to the crystal lattice.

In a standard synthesis protocol, a specific stoichiometric ratio of ligands such as oleic acid and oleylamine is utilized to ensure uniform growth. This study investigates the deviations in nanocrystal evolution when this thermodynamic control is intentionally compromised. Specifically, we examine the influence of halving the oleic acid concentration during the preparation of the cadmium precursor. Oleic acid binds to cadmium to form cadmium oleate; by reducing its concentration, the steric barrier shielding the growing QD surface is diminished. We hypothesize that this ligand deficiency will result in drastically accelerated growth kinetics, observable via a

rapid spectral red-shift, but at the cost of thermodynamic size control, leading to a broadened size distribution (Ostwald ripening) and reduced colloidal stability.

2 Experimental Details

2.1 Precursor Preparation and QD Synthesis

The synthesis of CdSe quantum dots was conducted using a hot-injection method [5]. A selenium stock solution was prepared by dissolving 99 mg of selenium powder in 5.5 mL of trioctylphosphine (TOP) and stirring at room temperature until the metallic selenium was fully dissolved [5]. The cadmium stock solution was prepared by mixing 53 mg of cadmium acetate dihydrate ($CdAc_2{\cdot}2H_2O$) with 5.5 mL of 1-octadecene [5]. To test our hypothesis regarding kinetic growth constraints, the concentration of the steric stabilizer was intentionally reduced; only 0.3 mL of oleic acid was added to the cadmium precursor, exactly half of the standard 0.6 mL parameter [5]. The cadmium mixture was heated to 130°C under continuous stirring until the precursor dissolved [5].

The primary reaction vessel, containing 10 mL of 1-octadecene and 0.6 mL of oleylamine, was heated to 165°C [5]. Upon reaching the target temperature, the selenium and cadmium stock solutions were sequentially injected [5]. To monitor the temporal evolution of the QDs, 0.5 mL aliquots were extracted from the reaction vessel at predefined time intervals and immediately quenched in a water bath to halt nanocrystal growth [5].

2.2 Nanocrystal Separation

To prepare the samples for high-resolution imaging and remove excess unreacted precursors and organic ligands, an aggressive anti-solvent precipitation protocol was employed [5]. For each 0.5 mL QD aliquot, a 3 mL anti-solvent mixture consisting of equal parts methanol, butanol, and acetone (1 mL each) was added [5]. The solution was centrifuged at 3300 RPM for 5 minutes, resulting in the precipitation of the QDs [5]. The supernatant was discarded, and the isolated QD precipitate was redispersed in 1 mL of non-polar hexane to achieve colloidal suspension [5].

2.3 Optical and Structural Characterization

The optical properties of the unwashed QD aliquots were characterized using UV-Vis absorption and fluorescence emission spectroscopy to determine the first exciton absorption peak and the emission full-width at half-maximum (FWHM), respectively. Structural characterization was performed using a Scanning Electron Microscope (SEM) operated in Scanning Transmission Electron Microscopy (STEM) mode at 30 keV [6]. The washed QD samples dissolved in hexane were drop-cast onto 400-mesh holey carbon TEM grids and dried under vacuum prior to imaging [6]. Image processing and physical sizing were conducted using ImageJ software [7].

3 Results and Discussion

The synthesis of CdSe quantum dots (QDs) showed a clear evolution in optical properties as the reaction progressed. UV-Vis absorption spectroscopy (Figure 1) revealed the first excitonic absorption peak continuously red-shifted from the blue/UV region toward higher wavelengths over 592.24 s, reflecting a reduction in band gap energy as particle size increased. This red shift reflects a reduction in band gap energy as the particle size increased.

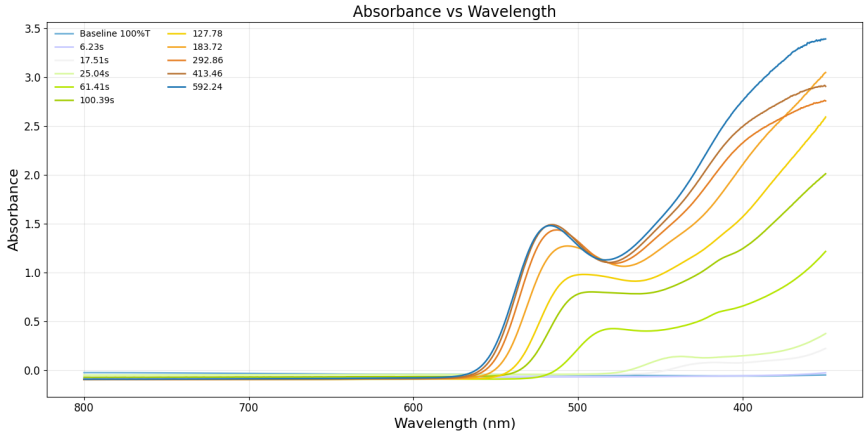


Figure 1: UV-Vis absorption spectra of CdSe quantum dots showing the progressive red shift of the excitonic absorption peak with increasing reaction time.

Normalized fluorescence emission spectra (Figure 2) similarly red-shifted over time, correlating with a visible color transition from blue/green to orange/red. The separated spectra also reveal changes in peak width and peak shape throughout the synthesis. Earlier samples exhibited broader and less defined emission features, while intermediate samples showed sharper and more intense peaks, indicating improved size uniformity during nanocrystal growth. At longer reaction times, slight peak broadening became visible again, suggesting an increase in particle size dispersion as growth continued.

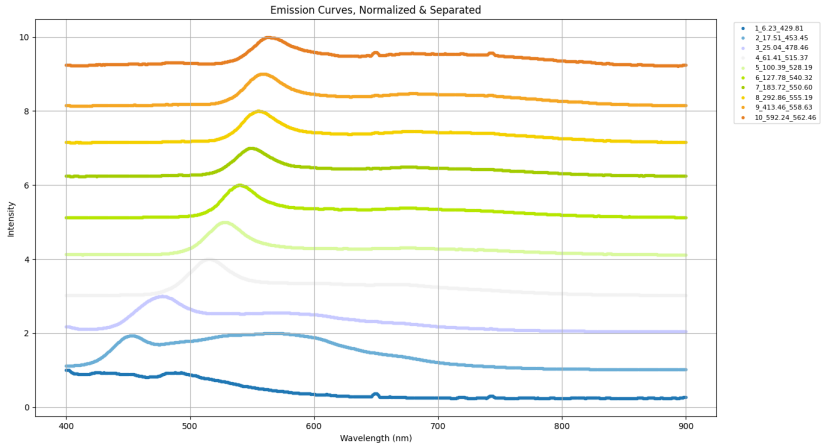


Figure 2: Separated fluorescence emission spectra of CdSe quantum dots showing the progressive red shift in emission wavelength and evolution of peak shape with increasing reaction time.

Emission peak broadening was quantified using the full-width at half-maximum (FWHM) of the normalized fluorescence spectra (Table 1; see Appendix B for detailed formulations). Intermediate reaction stages produced narrower emission peaks, while later-stage samples displayed broader peaks. This increase in FWHM at longer reaction times suggests that the size distribution became slightly broader as the reaction continued, likely due to reduced ligand passivation and increased variation in nanocrystal growth rates.

Physical samples corroborated these spectroscopic trends, transitioning from transparent to

Table 1: FWHM calculation from normalized emission spectra for the 0.3 mL oleic acid CdSe quantum dot samples.

Time (s)	$\boldsymbol{\lambda_1}$ **(nm)**	$\boldsymbol{\lambda_2}$ **(nm)**	**FWHM (nm)**
61.41	495.26	542.48	47.22
100.39	509.07	552.82	43.75
127.78	528.23	567.79	39.56
292.86	537.40	584.99	47.59
413.46	539.19	590.87	51.68

yellow/orange under ambient light and exhibiting blue-green to orange-red fluorescence under UV illumination (Figure 3). These visual observations matched the spectroscopic measurements and further confirmed the size-dependent optical behavior of the synthesized CdSe QDs.

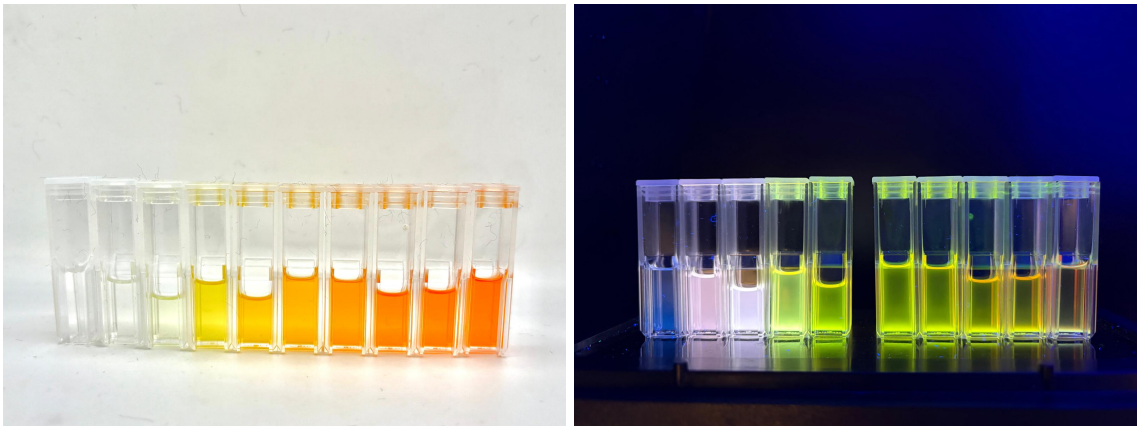


Figure 3: CdSe quantum dot samples under ambient light (left) and UV illumination (right), showing the evolution of visible color and fluorescence emission with increasing reaction time.

The relationship between the absorption and emission energies was analyzed using the Stokes shift. The Stokes shift ($\lambda_{\text{emission}} - \lambda_{\text{absorption}}$) increased from approximately 5.5 nm for the earliest sample to 43.4 nm at 592.24 s (Figure 4), suggesting energy relaxation became more significant as the quantum dots grew. Larger Stokes shifts may also indicate increased surface-state interactions and broader electronic relaxation pathways in the larger nanocrystals.

The diameter of the quantum dots was estimated from the absorption peak wavelength using the Peter Reiss empirical equation:

$$D = (1.6122 \times 10^{-9})\lambda^4 - (2.6575 \times 10^{-6})\lambda^3 + (1.6242 \times 10^{-3})\lambda^2 - 0.4277\lambda + 41.57 \qquad (1)$$

where D is the quantum dot diameter in nm and λ is the absorption peak wavelength in nm. The calculated diameter increased rapidly during the initial stage of the synthesis, rising from approximately 1.76 nm to 2.36 nm within the first 100 s. After this stage, the growth rate slowed considerably and gradually approached a plateau near 2.55 nm after approximately 600 s, as shown in Figure 5. This behavior suggests that rapid monomer attachment dominated the early stage of the reaction, while later growth became increasingly limited by precursor depletion and surface stabilization effects.

The observed growth behavior reflects the role of oleic acid as a surface-passivating ligand. Oleic acid and oleylamine bind to the nanocrystal surfaces and regulate the rate at which cadmium

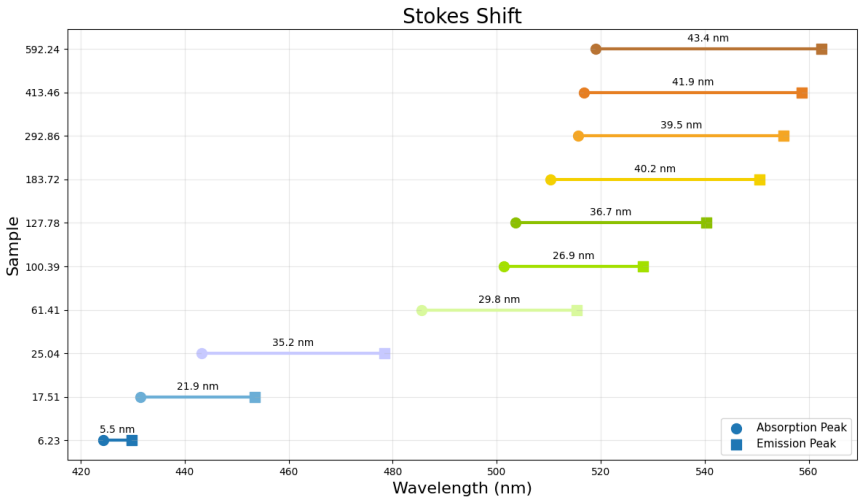


Figure 4: Stokes shift calculated from the difference between absorption and emission peak wavelengths for the 0.3 mL oleic acid CdSe quantum dot samples.

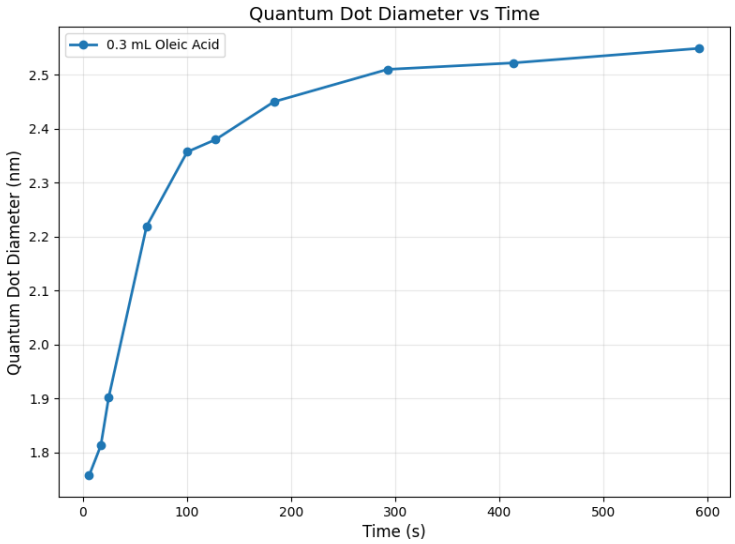


Figure 5: Quantum dot diameter as a function of reaction time for the 0.3 mL oleic acid CdSe quantum dot synthesis, calculated using the Peter Reiss empirical equation.

and selenium atoms attach to the growing lattice. In this synthesis, the oleic acid concentration was intentionally reduced to 0.3 mL, lowering the steric constraints on the system and allowing faster initial growth. However, the reduced ligand concentration may also have contributed to the broader FWHM values observed at later reaction times, indicating slightly reduced control over particle size uniformity.

Structural characterization was performed using STEM at 30 keV. The STEM image confirmed that the synthesized particles were within the nanometer size range and provided direct physical measurement of the nanocrystal dimensions. A particle diameter of approximately 3.75 nm was measured from the STEM micrograph.

For the final aliquot, the experimentally measured STEM diameter was 3.75 nm, yielding a 1.20

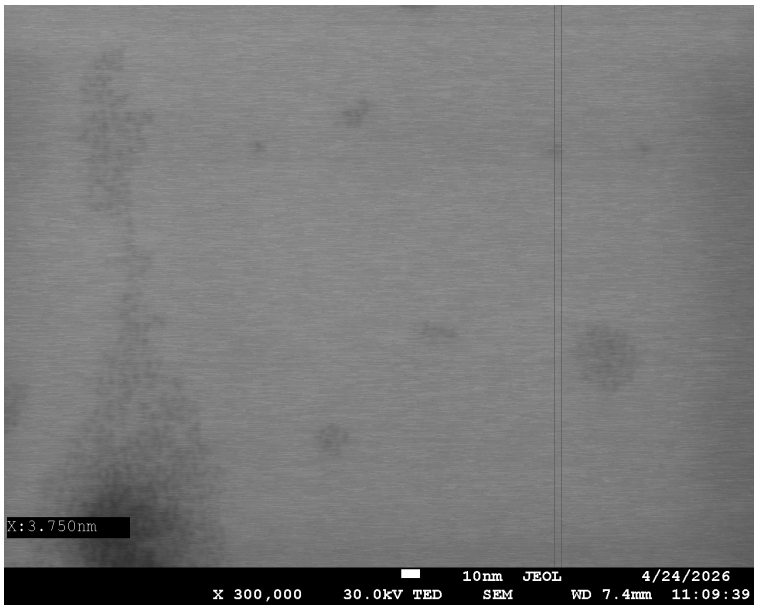


Figure 6: STEM image of CdSe quantum dots obtained at 30 keV, showing a measured particle diameter of approximately 3.75 nm.

nm deviation from the optically predicted diameter of 2.55 nm (see Appendix B for mathematical definitions). This difference may arise from uncertainties in peak identification, the presence of size dispersion within the sample, ligand-shell contributions, and the fact that the Peter Reiss equation estimates the optical diameter rather than directly measuring the physical particle size.

The optical trends observed throughout the synthesis are consistent with quantum confinement effects. Since the synthesized CdSe quantum dots were significantly smaller than the CdSe exciton Bohr radius of approximately 5.4 nm, the electron and hole became confined within a nanoscale volume. This confinement causes the electronic energy levels to become discrete rather than continuous, increasing the effective band gap for smaller particles. As the nanocrystals increased in size during synthesis, the band gap decreased, producing the observed red shifts in both absorption and emission spectra. The size-dependent optical behavior observed in the absorption spectra, fluorescence spectra, FWHM analysis, Stokes shift measurements, diameter calculations, and STEM characterization collectively confirms the successful synthesis of quantum-confined CdSe nanocrystals with tunable optical properties.

4 Conclusions

Ultimately, this study highlights the delicate balance between reaction speed and thermodynamic stability in quantum dot synthesis. By halving the oleic acid to 0.3mL, we successfully lowered the steric barrier and accelerated early stage growth, which was clearly evident in the rapid spectral red-shifting. However, this kinetic advantage came at a steep thermodynamic cost. Starving the reaction of its capping ligands led to severe colloidal instability, triggering Ostwald ripening quantified by the broadened emission spectra and causing the nanocrystals to physically aggregate under STEM imaging. These results demonstrate that while modulating ligands is a powerful way to control growth rates, a strict minimum threshold is absolutely essential to keep the quantum dots stable, separate, and uniform.

References

[1] Boatman, E. M., Lisensky, G. C., & Nordell, K. J. (2005). A safer, easier, faster synthesis for CdSe quantum dot nanocrystals. *Journal of Chemical Education*, 82(11), 1697.

[2] Landry, M. L., Morrell, T. E., Karagounis, T. K., Hsia, C. H., & Wang, C. Y. (2014). Simple syntheses of CdSe quantum dots. *Journal of Chemical Education*, 91(2), 274-279.

[3] Kagan, C. R., Lifshitz, E., Sargent, E. H., & Talapin, D. V. (2016). Building devices from colloidal quantum dots. *Science*, 353(6302), aac5523.

[4] Reiss, P., Protière, M., & Li, L. (2009). Core/shell semiconductor nanocrystals. *Small*, 5(2), 154-168.

[5] Kim, G. (2026). ESE 5360 - QD Lab: Synthesis and Separation. Singh Center for Nanotechnology, University of Pennsylvania.

[6] Kim, G. (2026). Lecture 18: FIB and TEM. ESE 5360. Singh Center for Nanotechnology, University of Pennsylvania.

[7] Kim, G. (2026). Lecture 17: Quantum Dot. ESE 5360. Singh Center for Nanotechnology, University of Pennsylvania.

I Appendix

A Data Tables

Table 2: Emission peak wavelength with respect to reaction time for the 0.3 mL oleic acid.

Sample	Time (s)	Emission Peak (nm)
1	6.23	429.81
2	17.51	453.45
3	25.04	478.46
4	61.41	515.37
5	100.39	528.19
6	127.78	540.32
7	183.72	550.60
8	292.86	555.19
9	413.46	558.63
10	592.24	562.40

B Calculation Details

The following formulations detail the calculations presented in the Results and Discussion section.

Full-Width at Half-Maximum (FWHM)

The full-width at half-maximum (FWHM) was utilized to quantify the emission peak broadening associated with Ostwald ripening. It is calculated by taking the difference between the higher

Table 3: Absorption peak wavelength as a function of sample number and reaction time for the 0.3 mL oleic acid CdSe quantum dot sample.

Sample	Time (s)	Absorption Peak (nm)
1	6.23	424.30
2	17.51	431.50
3	25.04	443.30
4	61.41	485.60
5	100.39	501.30
6	127.78	503.60
7	183.72	510.40
8	292.86	515.70
9	413.46	516.75
10	592.24	518.95

Table 4: Peter Reiss diameter calculation using absorption peak wavelengths for the 0.3 mL oleic acid CdSe quantum dot sample.

Sample	Time (s)	Absorption Peak (nm)	Diameter (nm)
1	6.23	424.30	1.757
2	17.51	431.50	1.813
3	25.04	443.30	1.902
4	61.41	485.60	2.219
5	100.39	501.30	2.357
6	127.78	503.60	2.380
7	183.72	510.40	2.450
8	292.86	515.70	2.510
9	413.46	516.75	2.522
10	592.24	518.95	2.549

Table 5: Stokes shift calculated from absorption and emission peak wavelengths for the 0.3 mL oleic acid CdSe quantum dot sample.

Sample	Time (s)	Abs Peak (nm)	Em Peak (nm)	Stokes Shift (nm)
1	6.23	424.30	429.81	5.51
2	17.51	431.50	453.45	21.95
3	25.04	443.30	478.46	35.16
4	61.41	485.60	515.37	29.77
5	100.39	501.30	528.19	26.89
6	127.78	503.60	540.32	36.72
7	183.72	510.40	550.60	40.20
8	292.86	515.70	555.19	39.49
9	413.46	516.75	558.63	41.88
10	592.24	518.95	562.40	43.45

wavelength (λ_2) and the lower wavelength (λ_1) where the normalized emission intensity is exactly half of its maximum value:

$$\text{FWHM} = \lambda_2 - \lambda_1 \tag{2}$$

Stokes Shift

The Stokes shift ($\Delta\lambda$) quantifies the energy lost to non-radiative relaxation processes prior to photon emission. It is defined as the difference between the peak emission wavelength and the first exciton absorption peak wavelength:

$$\Delta\lambda = \lambda_{\text{emission}} - \lambda_{\text{absorption}} \tag{3}$$

Theoretical vs. Experimental Sizing Deviation

To evaluate the accuracy of the empirical sizing model against the physical STEM characterization, the sizing deviation (ΔD) was determined by comparing the experimentally measured diameter (D_{STEM}) with the theoretically calculated diameter ($D_{\text{calculated}}$) derived from the Peter Reiss empirical equation for the final reaction aliquot:

$$\Delta D = D_{\text{STEM}} - D_{\text{calculated}} = 3.75 \text{ nm} - 2.55 \text{ nm} = 1.20 \text{ nm} \tag{4}$$

C Travelers of the group

ESE 5360 – QD Lab

Group B
Dhruvi Rathore
Krishna Madhur

1. Selenium stock solution preparation
    a. Weigh 99mg of Se powder and add to the solution.
    b. Add stirbar and 5.5mL of Trioctylphosphine (TOP) to a 25mL flask. Use a syringe to transfer TOP to the flask.
    c. Set the stirrer at 800 - 1000 rpm and stir the solution at room temperature until metallic Se is dissolved.
2. Cadmium stock solution preparation
    a. Weigh and add 53 mg of Cadmium Acetate Dihydrate ($CdAc_2 \cdot 2H_2O$). Add 0.6 ml of Oleic acid. → 0.3 ml oleic acid → Red shift.
    b. Add stirbar and 5.5mL of Octadecene to a 25mL flask.
    c. Connect the thermometer or thermocouple and place the assembly on the hotplate. Use stands and clamps to secure the system.
    d. Set the temperature of the hotplate to 130°C and the stirrer at 800 - 1000 rpm. Stir the solution until $CdAc_2 \cdot 2H_2O$ is dissolved.
3. QD solution preparation
    a. Add 10mL 1-octadecene and 0.6mL of Oleylamine to a 25mL flask and heat to 165°C.
4. Test tube setup
    a. Set up test tube rack and tubes in a water bath in order to quench the synthesis reaction. Arrange pipettes for quick access. Should be done in a couple of minutes.

5. QD solution withdrawal

   a. When the temperature reaches 165°C, add 2mL of Cadmium and Selenium precursor solution simultaneously using syringes and immediately begin timing

   b. Withdraw solution, about 1 ml at a time with pipettes, recording the time.

   c. Quickly put the solution into the test tubes sitting in the water bath. Keep track of which pipette is used at each time to prevent contamination later.

6. Suggested withdrawal times

   a. 8 sec (or sooner if possible!), 15 sec, 30sec, 1 min, 1.5 min, 2.5 min, 5 min, 7 min, 10 min. Keep a spare test tube handy in case the need arises.

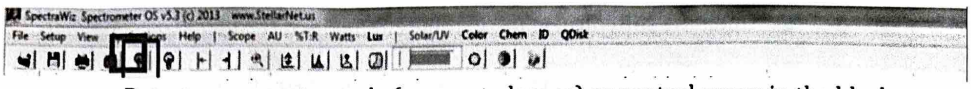
Sample #	Suggested Withdrawal Time	Actual Withdrawal Time (s)	Emission Peak Wavelength (nm)
1	8 sec	6.23	429.81
2	15 sec	17.51	453.45
3	30 sec	25.04	478.46
4	1 min	61.41	515.37
5	1.5 min	100.39	528.19
6	2 min	127.78	540.32
7	3 min	183.72	550.60
8	5 min	292.86	555.19
9	7 min	413.46	558.63
10	10min	592.24	562.40

7. Emission Data Collection

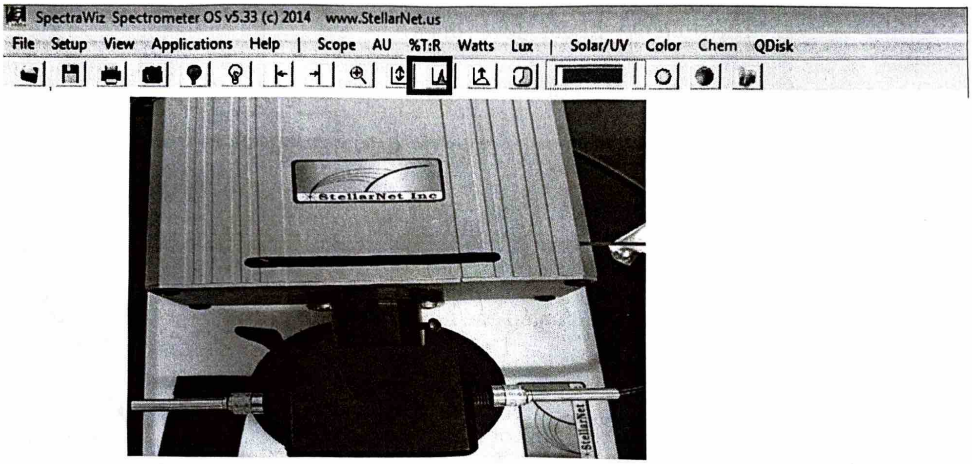


a. Open Software on the desktop.

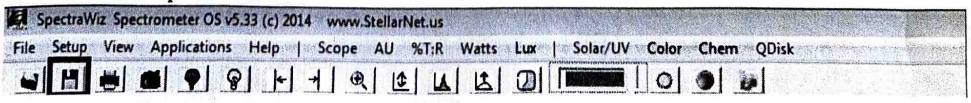
b. Click the icon with the dark light bulb. This takes a dark reference.



c. Put a transparent sample (pure octadecene) as control group in the black box. The software will automatically take the spectra. Click save button to save results.



d. Take out the transparent sample and put in another sample. Click the peak button on top and measure the peak of the curve. If there are more than one peaks, move the red line on the screen and measure the highest peak of the curve. Take down the emission peak wavelength in the table above and save the spectra data.



e. Put other samples in the black box one-by-one, measure each emission peak and save all data.

8. Absorption Data Collection

   a. Open the software on the computer and open the lid of the container.

   b. One student put on gloves and put a transparent sample in the container. The other student closes the lid and control the software to measure the absorption spectra.

   c. The student without gloves open and close the lid while the student with the gloves on change the sample. Measure the spectra again.

   d. After testing all samples, save the data and close the software. Take out the sample and clean the lab.

9. QD Separation

   a. Pour Butanol, Methanol and Acetone into three beakers (TA)

   b. Extract 0.5ml QDs solution from cuvettes using syringes

   c. In a test tube, add 1mL of Methanol, 1mL of Butanol and 1mL of Acetone to 0.5ml QD solution using syringes. Then, make turbulence with a pipette gently.

   d. Wrap the test tube top with Parafilm. Centrifuge solution at 3300 RPM for 5 minutes.

   e. Extract the precipitate by removing the clear solution using pipets. Be careful not to extract QD precipitate by accident.

   f. Dissolve QD precipitate in 1ml Hexane.

   g. Dispense the QDs on the TEM grid as instructed